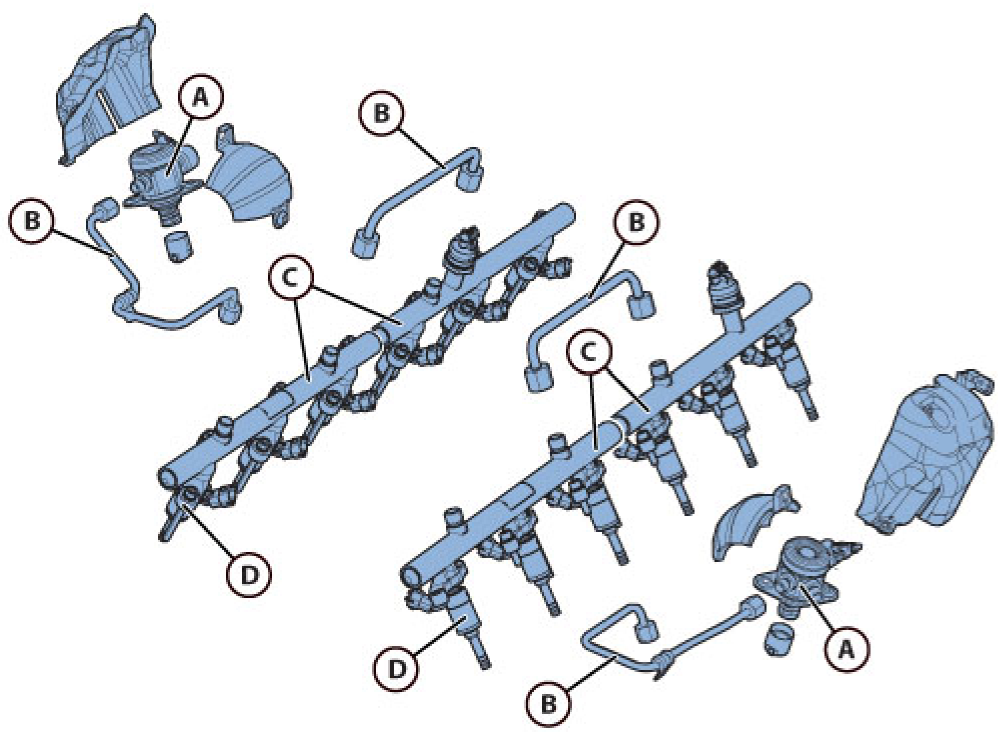
GTC4Lusso USA - GDI direct injection system - Latest Update: 2021-01-26



## F3.14 GDI direct injection system

### System layout




**A** - High pressure fuel pump
**B** - High pressure fuel pipes
**C** - Rail
**D** - Injector

### Removing the GDI injection pumps

**CAUTION**

Fuel is highly flammable.
NEVER WORK ON A HOT ENGINE.

**CAUTION**

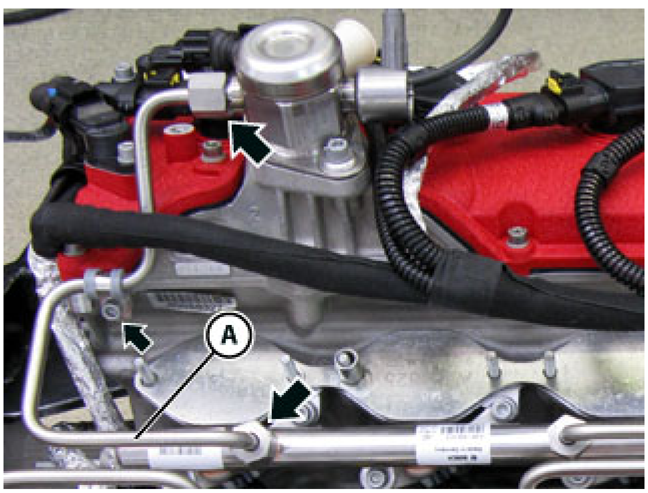
When connecting a fuel system pipe, bear in mind that the line may be pressurised. Proceed therefore with caution and keep away any type of flame as there is a risk of fuel spraying onto the operator.
Always plug the disconnected fuel pipes to prevent residual fuel from leaking.

**CAUTION**

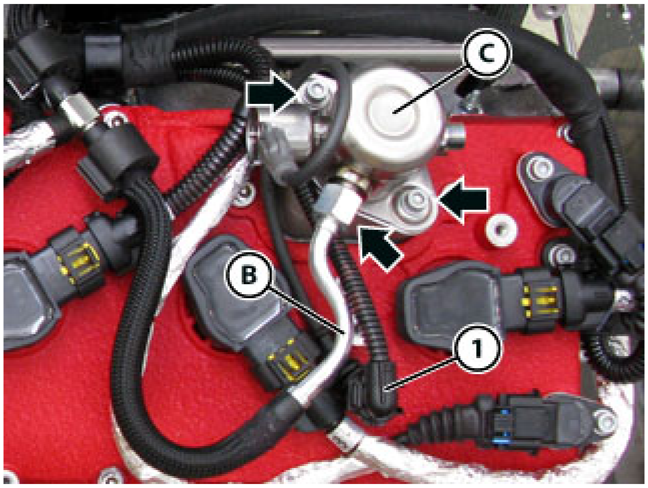
Before performing the next operation, cover the relevant part of the engine with rags so that any leaking fuel will be absorbed and will not come into contact with the engine or any other unit. In case of contact with fuel, dry the affected area immediately.

- Deactivate the Park Lock mechanically (E5.10).
- Connect the DEIS tester to the diagnostic socket (F2.09).
  - Perform the pressure release procedure for the high pressure circuit using the DEIS diagnostic tester.
- Disconnect the battery (F2.01).
- Remove the engine intake manifold (B4.04).

**LH GDI injection pump**




- Undo the indicated screw and retrieve the respective rubber lined bracket.
- Disconnect and remove the pipe **(A)**, undoing the respective unions indicated.




- Disconnect the connector **(1)**.
- Disconnect the pipe **(B)** undoing the respective union indicated.

The trilobe on the camshaft may be pushing against the pump.

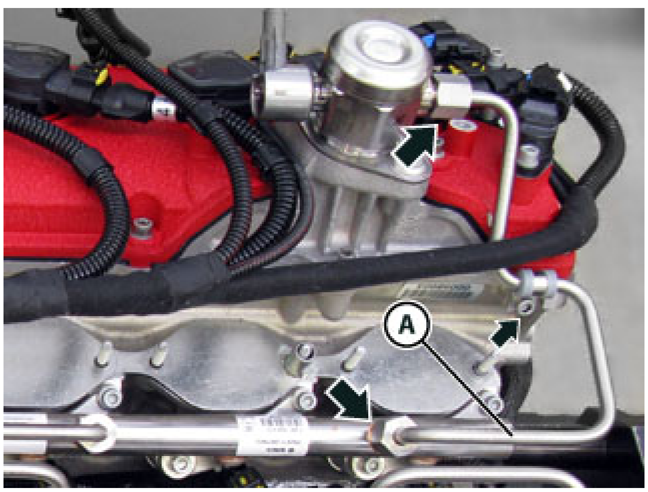
When removing the high pressure fuel pump fasteners, keep the pump pressed into its seat.

- Remove the pump **(C)**, undoing the indicated screws carefully.

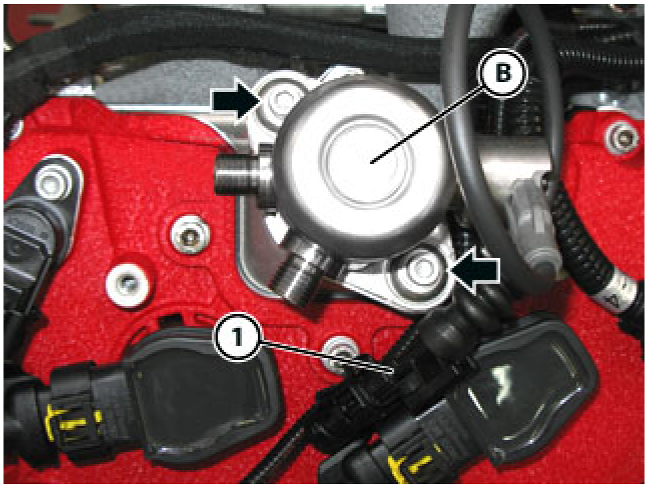

- Retrieve the cup **(D)**.

- Plug the fuel orifices on the pump and on the rail.

**RH GDI injection pump**




- Undo the indicated screw and retrieve the respective rubber lined bracket.
- Disconnect and remove the pipe **(A)**, undoing the respective unions indicated.




- Disconnect the connector **(1)**.

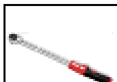
The trilobe on the camshaft may be pushing against the pump.
When removing the high pressure fuel pump fasteners, keep the pump pressed into its seat.

- Remove the pump **(B)**, undoing the indicated screws carefully.


- Retrieve the cup **(D)**.

- Plug the fuel orifices on the pump and on the rail.

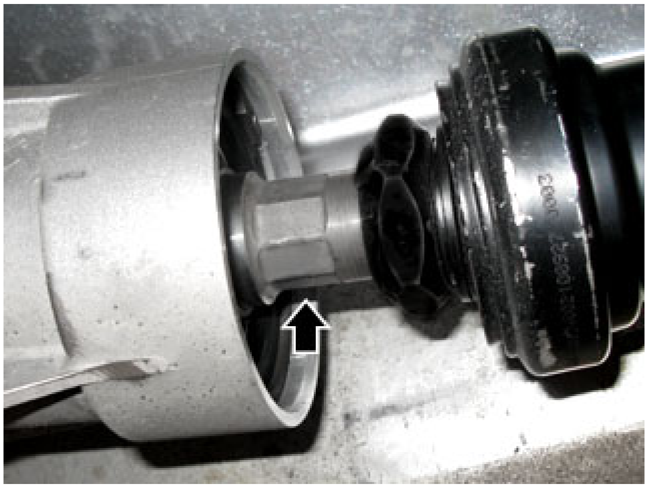
**Refitting the GDI injection pumps**

|  | | | |
|---|---|---|---|
| **Tightening torque** | | **Nm** | **Class** |
| Fastening high pressure fuel pump | Screw (pretightening) | 5 Nm | B |
| | Screw | 10 Nm | B |
| Fastening low pressure fuel pipe to high pressure fuel pumps | Union | 20 $^{+2}_{-0}$ Nm | 0 |
| Fastening the high pressure GDI fuel pipes | Union | 23 $^{+1}_{-1}$ Nm | 0 |
| Fastening the rubber-lined bracket holding the high pressure GDI fuel pipe | Screw | 8 Nm | A |

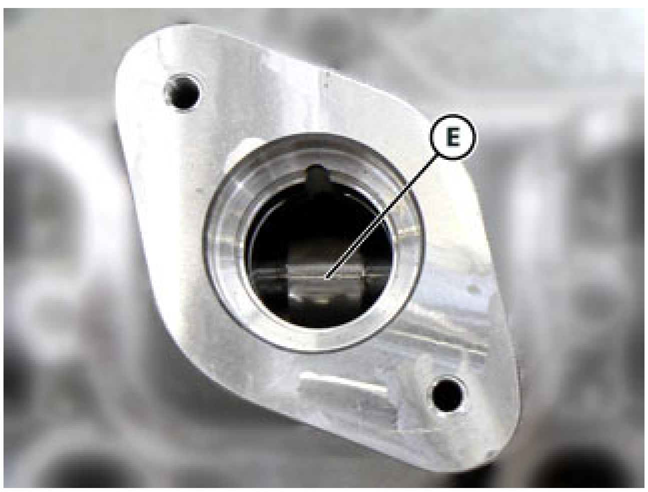
ENSURE THAT all the components are CLEAN before fitting.
ENSURE that the injector seats on the cylinder head are completely CLEAN.

- Remove the transmission shaft (C5.02).
  - Remove the transmission shaft heat shield.

### LH GDI injection pump



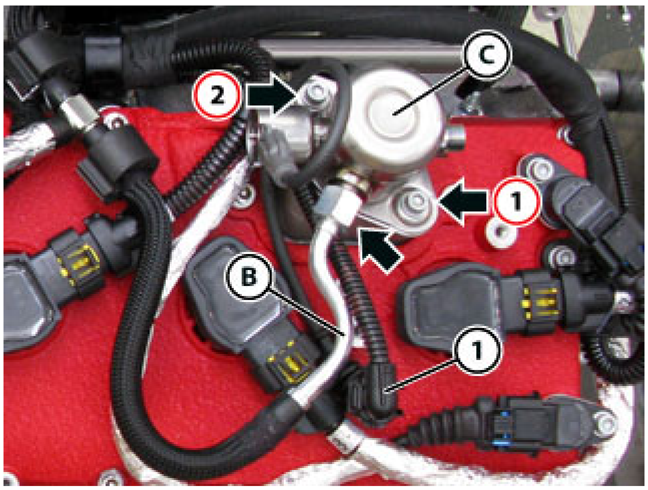
- Apply a suitable wrench to the point indicated on the transmission shaft in order to be able to turn the shaft manually.



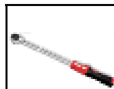

- Using the wrench fitted previously, manually turn the driveshaft to bring the intake camshaft into a position with the injection pump trilobe **(E)** at BDC.
  - In the position, the trilobe is not pushing against the injection pump.


- Fit the cup **(D)**, lubricating with the specified engine oil.
- Lubricate the seat of the fuel pump O-ring with the specified engine oil.

- Check that the O-ring indicated is installed correctly, and lubricate the O-ring with the specified engine oil.

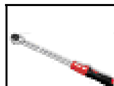



- Fit the pump **(C)** in the correct position and tighten the respective screws indicated.
- Pre-tighten and tighten the screws fastening the pump **(C)** in the sequence indicated.



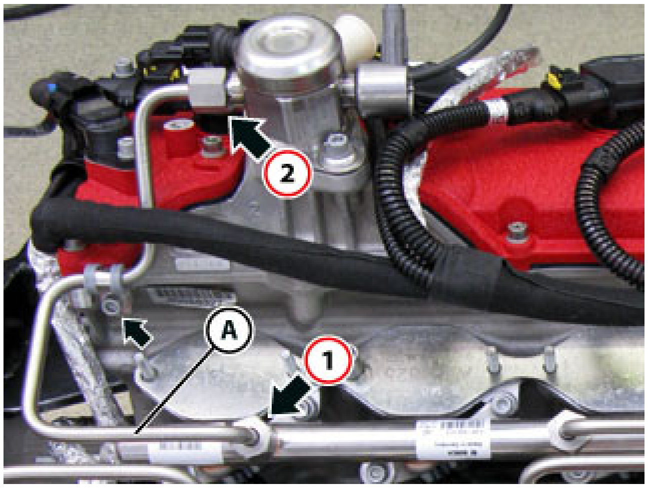
| **Tightening torque** | **Nm** | **Class** |
|---|---|---|
| Screw (pretightening) | 5 Nm | B |
| Screw | 10 Nm | B |

- Connect the pipe **(B)**, tightening the respective union indicated.



| **Tightening torque** | **Nm** | **Class** |
|---|---|---|
| Union | 20 $^{+2}_{-0}$ Nm | 0 |

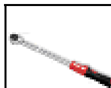
- Connect the connector **(1)**.




- Fit and connect the pipe **(A)**, lubricating and tightening the respective unions indicated
- Tighten the indicated screws fastening the pipe **(A)** in the sequence given.

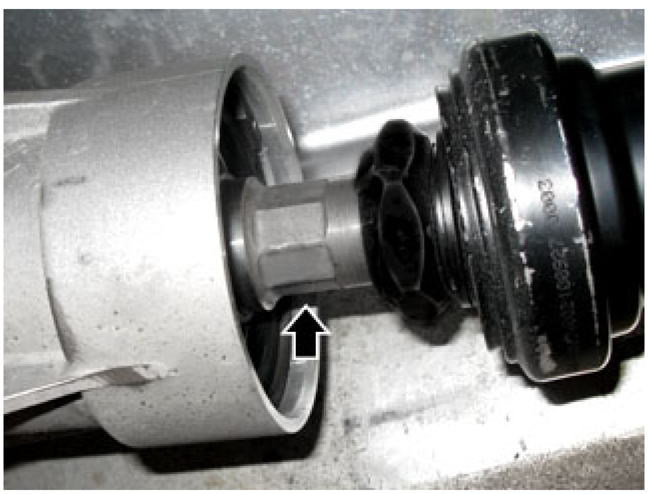
| Tightening torque | Nm | Class |
|---|---|---|
| Union | $23^{+1}_{-1}$ Nm | 0 |

- Tighten the indicated screw, fitting the relative rubber lined bracket.

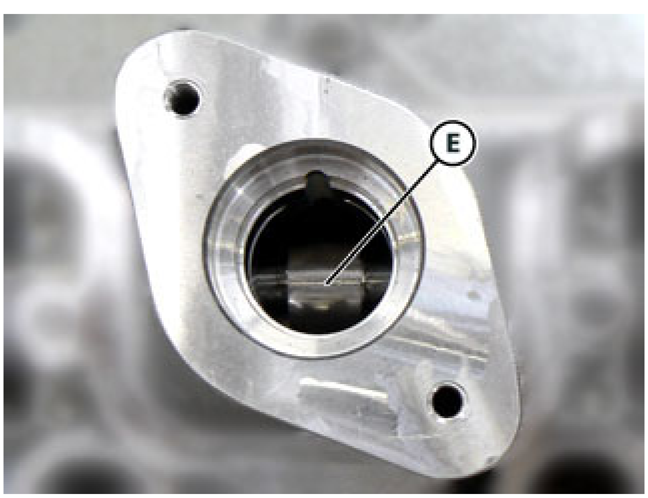


| Tightening torque | Nm | Class |
|---|---|---|
| Screw | 8 Nm | A |

**Right hand pump**



- Apply a suitable wrench to the point indicated on the transmission shaft in order to be able to turn the shaft manually.

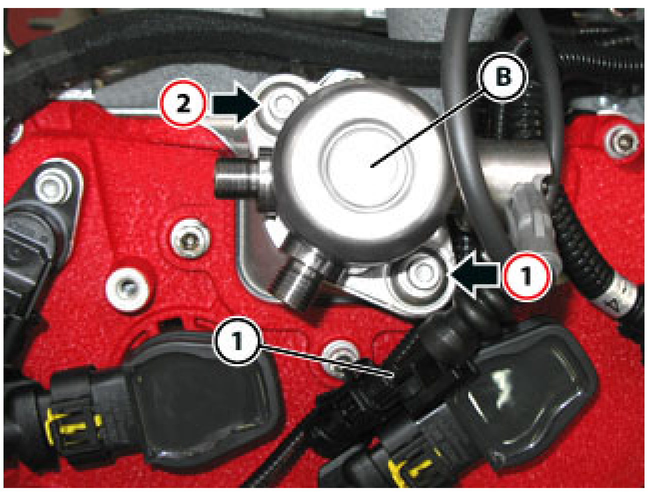



- Using the wrench fitted previously, manually turn the driveshaft to bring the intake camshaft into a position with the injection pump trilobe **(E)** at BDC.

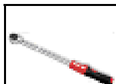
*i* In the position, the trilobe is not pushing against the injection pump.


- Fit the cup **(D)**, lubricating with the specified engine oil.
- Lubricate the seat of the fuel pump O-ring with the specified engine oil.

- Check that the O-ring indicated is installed correctly, and lubricate the O-ring with the specified engine oil.

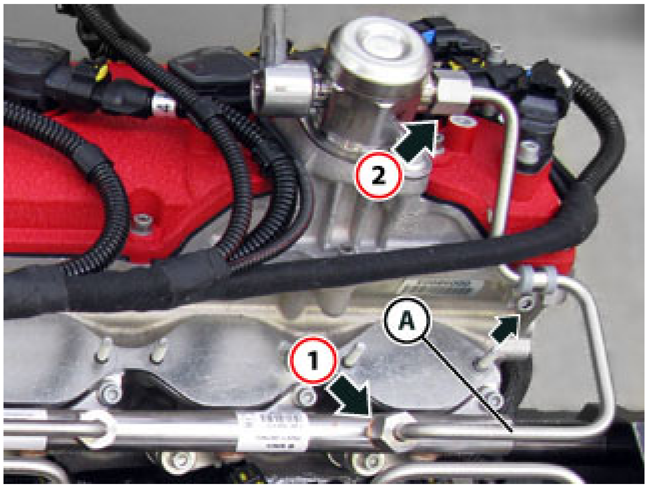



- Fit the pump **(B)** in the correct position and tighten the respective screws indicated.
- Pre-tighten and tighten the screws fastening the pump **(B)** in the sequence indicated.



| Tightening torque | Nm | Class |
|---|---|---|
| Screw (pretightening) | 5 Nm | B |
| Screw | 10 Nm | B |

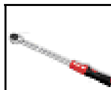
- Connect the connector **(1)**.




- Fit and connect the pipe **(A)**, lubricating and tightening the respective unions indicated
- Tighten the indicated screws fastening the pipe **(A)** in the sequence given.

|  | | |
|---|---|---|
| **Tightening torque** | **Nm** | **Class** |
| Union | 23 $^{+1}_{-1}$ Nm | 0 |

- Tighten the indicated screw, fitting the relative rubber lined bracket.

|  | | |
|---|---|---|
| **Tightening torque** | **Nm** | **Class** |
| Screw | 8 Nm | A |

➤ Refit the transmission shaft (C5.02).
  Refit the transmission shaft heat shield.

➤ Refit the intake manifold (B4.04).
➤ Connect the battery (F2.01).
➤ Deactivate the Park Lock mechanically (E5.10).
  Reactivate the Park Lock.

➤ Connect the DEIS tester to the diagnostic socket (F2.09).
  Perform the "High pressure circuit leakage test" with the DEIS diagnostic tester.

### Removing the GDI injection rail pipes

**CAUTION**
Fuel is highly flammable.
NEVER WORK ON A HOT ENGINE.

**CAUTION**
When connecting a fuel system pipe, bear in mind that the line may be pressurised. Proceed therefore with caution and keep away any type of flame as there is a risk of fuel spraying onto the operator.
Always plug the disconnected fuel pipes to prevent residual fuel from leaking.

**CAUTION**

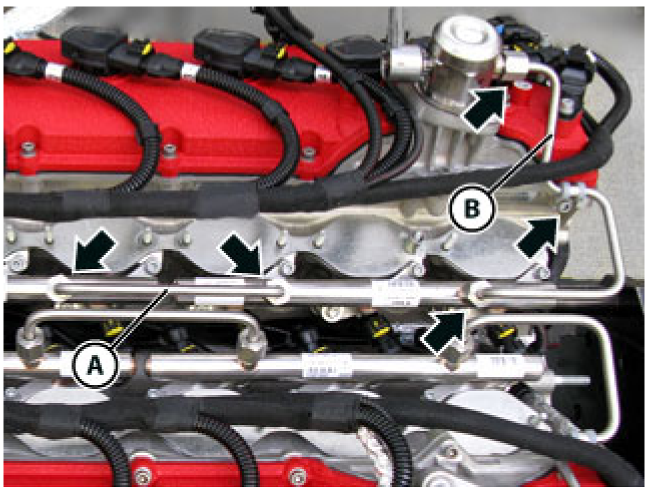
Before performing the next operation, cover the relevant part of the engine with rags so that any leaking fuel will be absorbed and will not come into contact with the engine or any other unit. In case of contact with fuel, dry the affected area immediately.

When replacing the GDI injector rail pipes, the respective connector pipes must also be removed following the instructions given in the Spare Parts Catalogue.

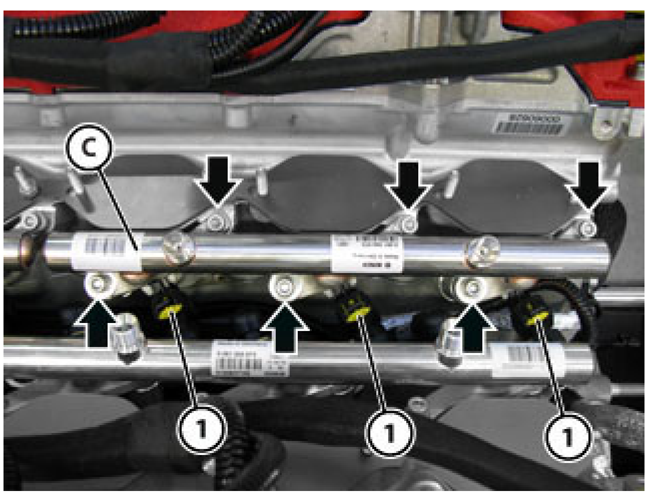
- Connect the DEIS tester to the diagnostic socket (F2.09).
  - Perform the pressure release procedure for the high pressure circuit using the DEIS diagnostic tester.

- Disconnect the battery (F2.01).
- Remove the engine intake manifold (B4.04).

**RH rear GDI injection rail pipe**

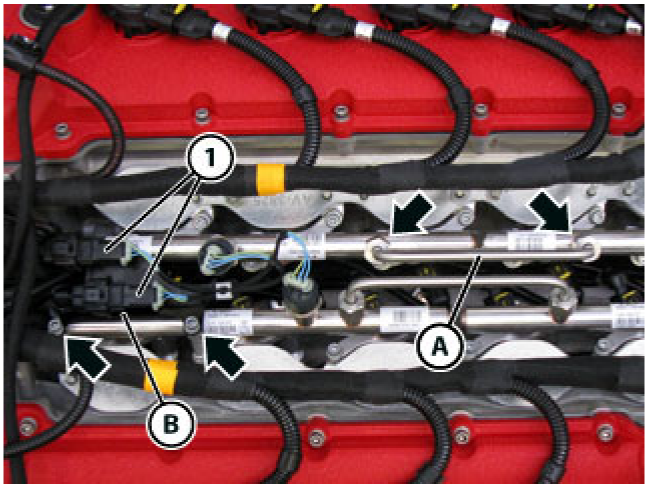



- Remove the pipe **(A)**, undoing the respective unions indicated.
- Remove the pipe **(B)** undoing the respective unions and the screw indicated.

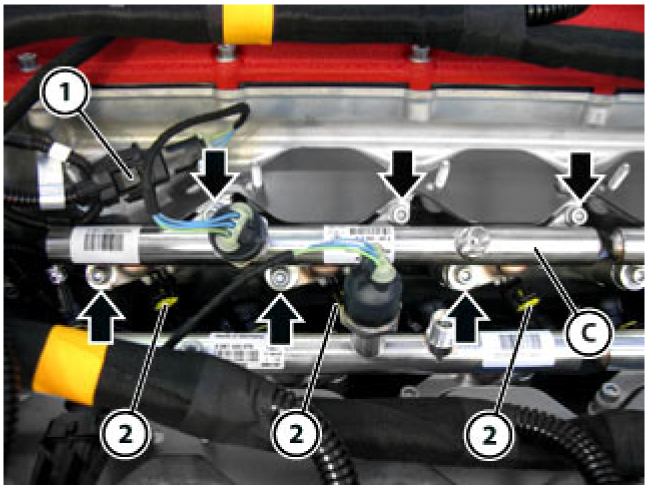



- Undo the screws indicated.
- Disconnect the connectors **(1)**.
- Remove the rail **(C)**, removing the relative injectors carefully.

**RH front GDI injection rail pipe**

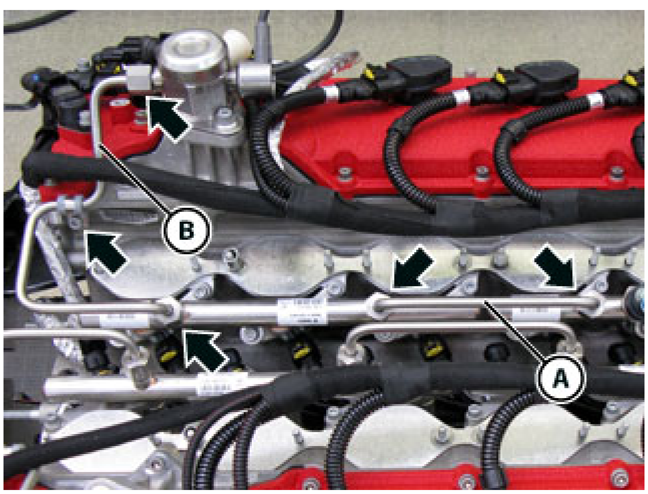



- Remove the pipe **(A)**, undoing the respective unions indicated.
- Remove the bracket **(B)**, if fitted, by undoing the respective screws indicated and disconnecting the connectors **(1)**.

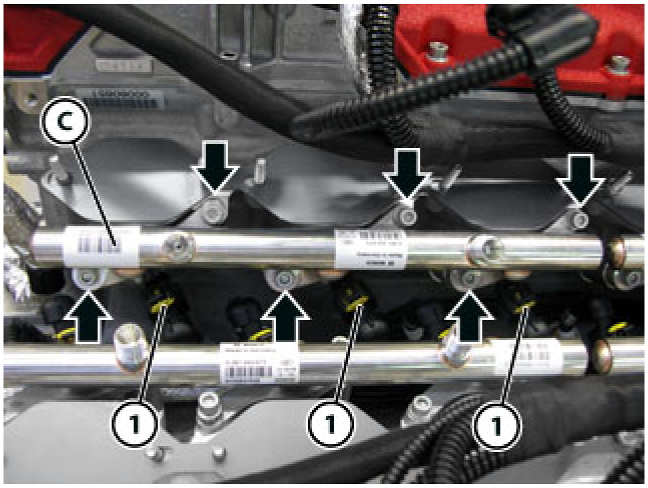



- Undo the screws indicated.
- Disconnect the connector **(1)**.
- Disconnect the connectors **(2)**.
- Remove the rail **(C)**, removing the relative injectors carefully.

**LH rear GDI injection rail pipe**

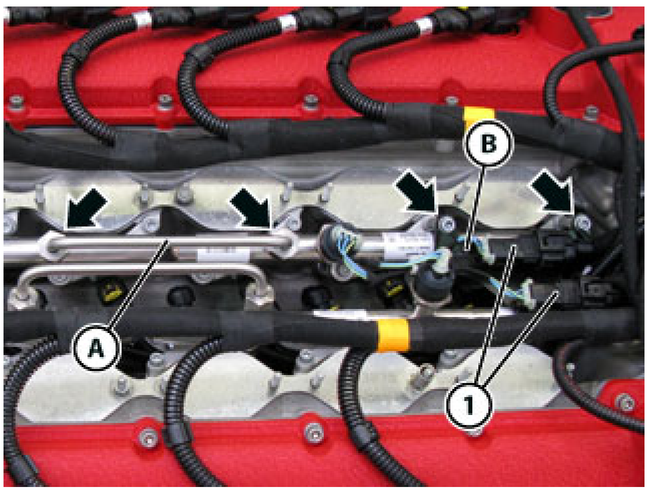



- Remove the pipe **(A)**, undoing the respective unions indicated.
- Remove the pipe **(B)** undoing the respective unions and the screw indicated.

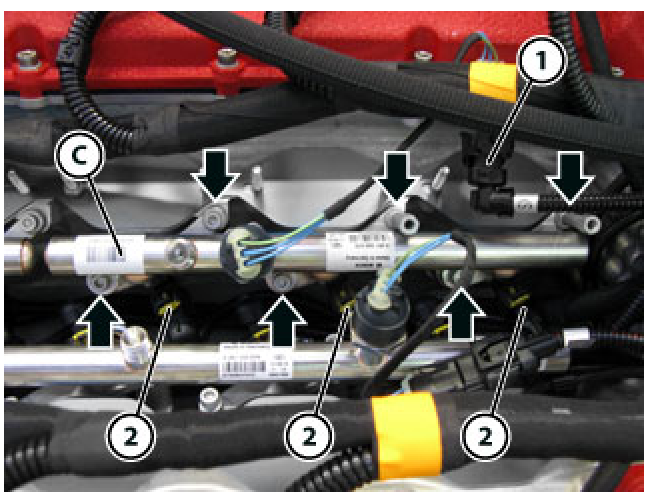



- Undo the screws indicated.
- Disconnect the connectors **(1)**.
- Remove the rail **(C)**, removing the relative injectors carefully.

**LH front GDI injection rail pipe**




- Remove the pipe **(A)**, undoing the respective unions indicated.
- Remove the bracket **(B)**, if fitted, by undoing the respective screws indicated and disconnecting the connectors **(1)**.




- Undo the screws indicated.
- Undo the indicated stud bolts.
- Disconnect the connector **(1)**.
- Disconnect the connectors **(2)**.
- Remove the rail **(C)**, removing the relative injectors carefully.

➢ Cover the injector seats on the cylinder head to prevent foreign objects from entering.

**Refitting the GDI injection rail pipes**

| Tightening torque | | Nm | Class |
|---|---|---|---|
| Fastening the GDI rail pipes | Screw (pretightening) | 5 Nm | B |
| | Screw | 10 Nm | B |
| Fastening the high pressure GDI fuel pipes | Union | 23 $^{+1}_{-1}$ Nm | 0 |
| Fastening the rubber-lined bracket holding the high pressure GDI fuel pipe | Screw | 8 Nm | A |
| Fastening the GDI rail pipe connector bracket | Screw | 10 Nm | B |

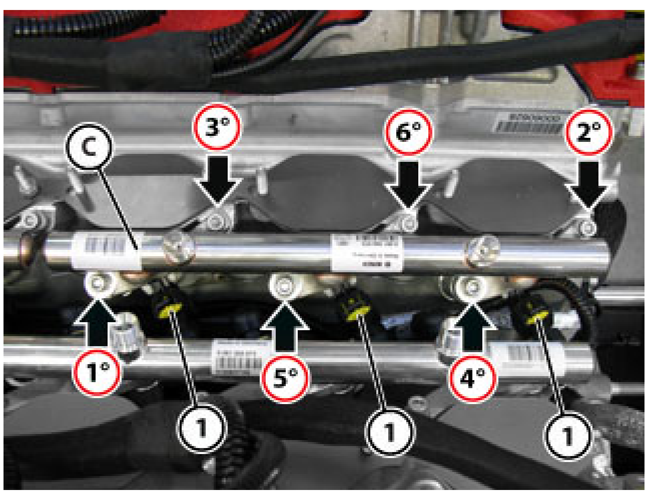
ENSURE THAT all the components are CLEAN before fitting.
ENSURE that the injector seats on the cylinder head are completely CLEAN.

NEVER lubricate the injector tip side Teflon seal rings.

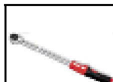
When replacing the GDI injector rail pipes, the respective connector pipes must also be removed following the instructions given in the Spare Parts Catalogue.

- Replace the retainer clips.
- Replace the Teflon seal rings and the O-rings.

**RH rear GDI injection rail pipe**

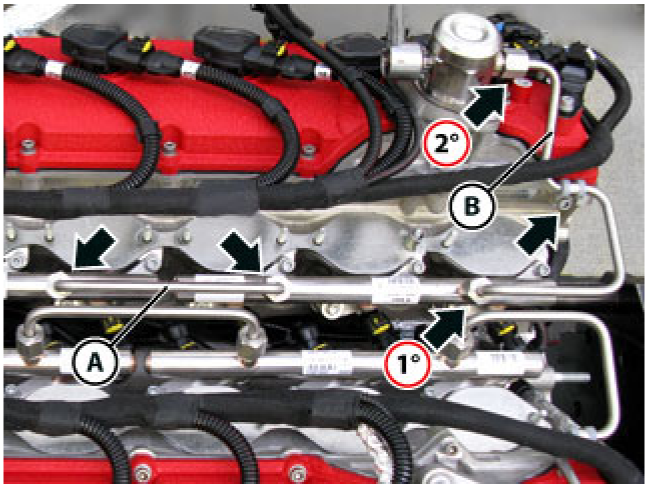



- Fit the rail pipe **(C)**, inserting the relative injectors carefully.
- Tighten the screws indicated.
- Tighten the indicated screws in the sequence given.



| Tightening torque | Nm | Class |
|---|---|---|
| Screw (pretightening) | 5 Nm | B |
| Screw | 10 Nm | B |

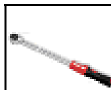
- Connect the connectors **(1)**.




- Fit the pipe **(B)**, lubricate and hand-tighten the respective unions indicated, then tighten in the sequence given.

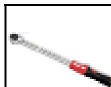
| Tightening torque | Nm | Class |
|---|---|---|
| Union | $23^{+1}_{-1}$ Nm | 0 |

- Tighten the indicated screw, fitting the relative rubber-lined bracket.



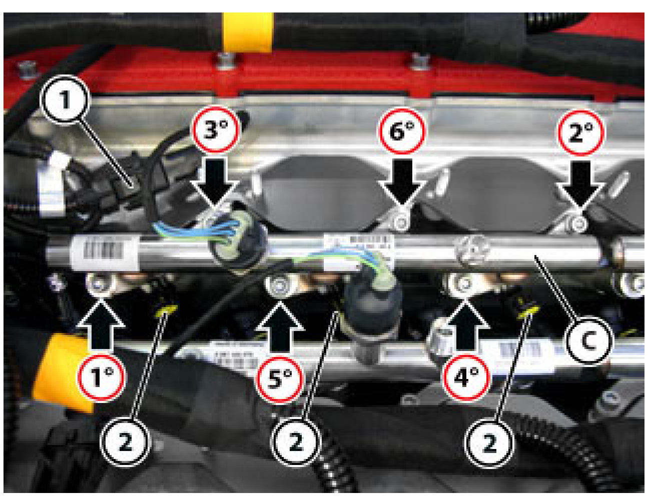
| Tightening torque | Nm | Class |
|---|---|---|
| Screw | 8 Nm | A |

- Fit the pipe **(A)**, then hand-tighten and subsequently tighten the respective unions indicated.

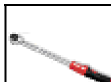


| Tightening torque | Nm | Class |
|---|---|---|
| Union | $23^{+1}_{-1}$ Nm | 0 |

**RH front GDI injection rail pipe**

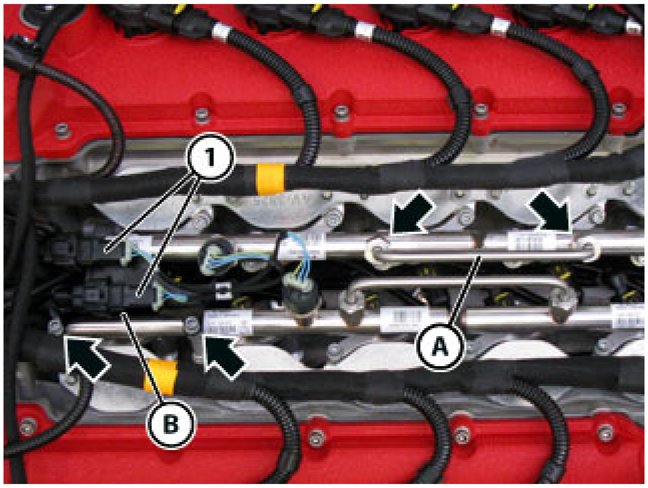



- Fit the rail pipe **(C)**, inserting the relative injectors carefully.
- Tighten the screws indicated.
- Tighten the indicated screws in the sequence given.



| Tightening torque | Nm | Class |
|---|---|---|
| Screw (pretightening) | 5 Nm | B |
| Screw | 10 Nm | B |

- Connect the connectors **(2)**.
- Connect the connector **(1)**.



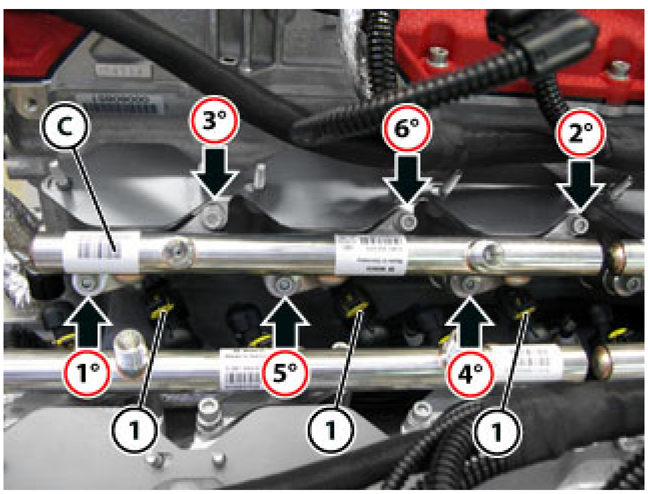

- Fit the bracket **(B)**, if applicable, tighten the respective screws indicated and connect the connectors **(1)**.

| **Tightening torque** | **Nm** | **Class** |
|---|---|---|
| Screw | 10 Nm | B |

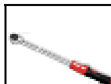
- Fit the pipe **(A)**, then hand-tighten and subsequently tighten the respective unions indicated.

| **Tightening torque** | **Nm** | **Class** |
|---|---|---|
| Union | $23^{+1}_{-1}$ Nm | 0 |

**LH rear GDI injection rail pipe**

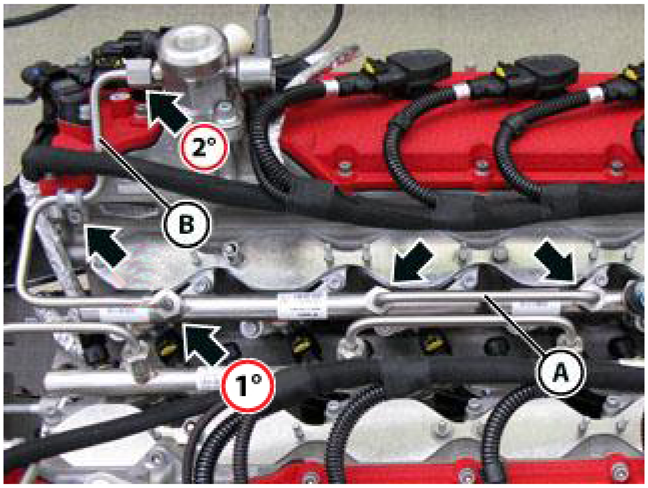



- Fit the rail pipe **(C)**, inserting the relative injectors carefully.
- Tighten the screws indicated.
- Tighten the indicated screws in the sequence given.



| **Tightening torque** | **Nm** | **Class** |
|---|---|---|
| Screw (pretightening) | 5 Nm | B |
| Screw | 10 Nm | B |

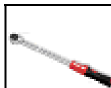
- Connect the connectors **(1)**.




- Fit the pipe **(B)**, lubricate and hand-tighten the respective unions indicated, then tighten in the sequence given.

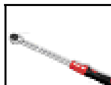
| Tightening torque | Nm | Class |
|---|---|---|
| Union | 23 $^{+1}_{-1}$ Nm | 0 |

- Tighten the indicated screw, fitting the relative rubber-lined bracket.



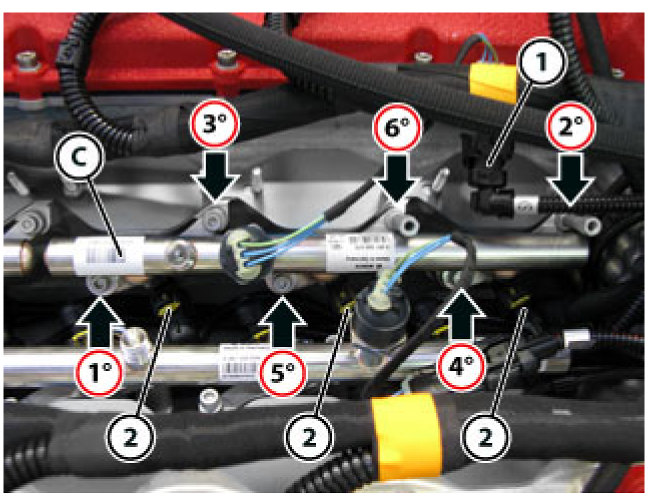
| Tightening torque | Nm | Class |
|---|---|---|
| Screw | 8 Nm | A |

- Fit the pipe **(A)**, then hand-tighten and subsequently tighten the respective unions indicated.



| Tightening torque | Nm | Class |
|---|---|---|
| Union | 23 $^{+1}_{-1}$ Nm | 0 |

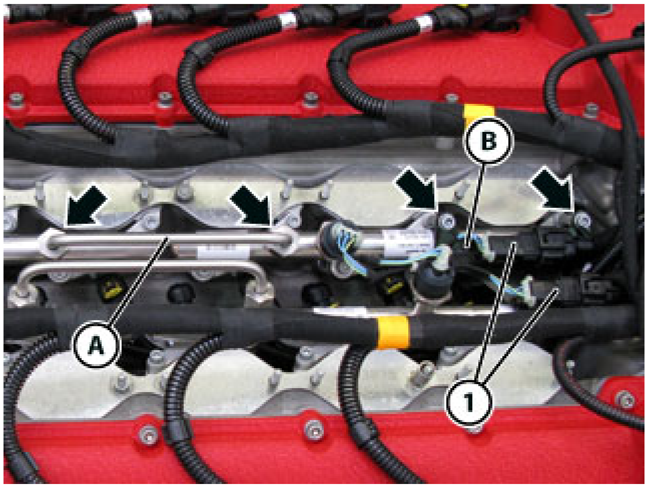
**LH front GDI injection rail pipe**




- Fit the rail pipe **(C)**, inserting the relative injectors carefully.
- Hand-tighten the indicated screws and stud bolts.
- Definitively tighten the indicated screws and stud-bolts in the sequence given.

| Tightening torque | Nm | Class |
|---|---|---|
| Screw (pretightening) | 5 Nm | B |
| Screw | 10 Nm | B |

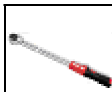
- Connect the connectors **(2)**.
- Connect the connector **(1)**.




- Fit the bracket **(B)**, if applicable, tighten the respective screws indicated and connect the connectors **(1)**.

| Tightening torque | Nm | Class |
|---|---|---|
| Screw | 10 Nm | B |

- Fit the pipe **(A)**, then hand-tighten and subsequently tighten the respective unions indicated.



| Tightening torque | Nm | Class |
|---|---|---|
| Union | $23^{+1}_{-1}$ Nm | 0 |

- Refit the intake manifold ( B4.04).
- Connect the battery ( F2.01).
- Connect the DEIS tester to the diagnostic socket ( F2.09).
  - Perform the "High pressure circuit leakage test" with the DEIS diagnostic tester.